Measuring Economic Risk in Africa: The African Economic Risk Index and African Risk Score

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Abstract

We construct news-based measures of economic risk for 54 African economies from 2015 to 2025. The African Economic Risk Index (AERI) assigns graded scores to articles according to economic relevance, severity and textual emphasis and averages those scores over the monthly news flow. Eight non-exclusive pillar indices retain the domain information. The African Risk Score (ARS) maps bounded measures of AERI intensity, relevant-article breadth, average severity and active-domain count to a 0–100 scale. The data contain 21.30 million retained article records and 6,488 country-months that meet the principal descriptive coverage threshold. Exact-text deduplication removes 497,091 repeated records (2.33 percent) from the reconstructed frozen vintage; original and duplicate-adjusted AERI remain highly correlated (Pearson 0.9296; rank 0.9883). Aggregate AERI peaks in April 2020, but country histories differ in timing and composition. Alternative ARS weights produce closely aligned scores. In pooled panel VARs, inflation-risk readings precede higher headline inflation, exchange-rate depreciation is followed by higher AERI, and stronger real activity is followed by lower AERI. An external monetary-policy design also shows larger subsequent AERI increases after U.S. tightening shocks in economies with greater predetermined external-debt exposure. The reported sensitivity and falsification checks do not materially change these findings. AERI, the pillar indices and ARS are decomposable measures of reported economic risk that can be read alongside conventional macroeconomic statistics.

Keywords: Economic risk; Africa; newspapers; text as data; risk measurement; alternative data.

JEL Classification: E37, C55, C81, O55.

1 Introduction

Official statistics remain the main source of information about inflation, output, fiscal conditions and external balances, but they do not capture all changes in the economic environment as they occur. News reports often record shortages, financing pressures, currency stress, price increases and production disruptions before the full effects appear in conventional data. We develop a way to measure that information systematically across African economies.

We construct the **African Economic Risk Index (AERI)** from article-level scores that combine economic relevance, severity and textual emphasis. For each country and month, AERI averages these scores over all retained articles and multiplies the result by 100. Non-relevant articles remain in the denominator, so the index rises when economic-risk reporting becomes more prevalent, more severe or more prominent relative to the overall news flow.

We also construct eight non-exclusive pillar indices covering inflation, foreign-exchange and external conditions, debt and fiscal conditions, the financial sector, growth and the real economy, commodities and supply, political economy, and climate and disasters. These indices show where measured risk is concentrated. A separate PCA summary records common movement across the pillars. Finally, the **African Risk Score (ARS)** combines AERI intensity, breadth of relevant reporting, average severity and the number of active domains on a 0–100 scale. AERI is the principal measure; the pillars and ARS provide additional information about the composition and breadth of the underlying news signal.

The data contain 21.30 million retained article records for 54 African economies from January 2015 through December 2025. We observe 7,127 country-months, of which 6,488 contain at least 100 retained articles and form the principal descriptive sample. Aggregate AERI reaches its sample maximum in April 2020. The national series do not peak at the same time, and their risk composition differs across countries. ARS is also insensitive to an alternative weighting rule: the benchmark and PCA-weighted versions have a correlation of 0.995 in levels and 0.997 in ranks.

To examine economic content, we estimate pooled panel vector autoregressions (PVARs) linking the news measures to inflation, exchange-rate movements and monthly GDP growth. Inflation-risk lags foreshadow higher inflation, depreciation is followed by higher AERI, and stronger real activity is followed by lower AERI. We also use high-frequency U.S. monetary-policy surprises from Jarociński and Karadi (2020) as an external source of variation and interact them with 2014 external-debt exposure. AERI rises more after a tightening in African economies that entered the sample with greater external exposure. The differential response remains under alternative exposure measures, an alternative monetary-policy shock, leave-one-country-out estimation, and a sample restricted to FOMC announcement months.

The construction builds on the newspaper-based measurement approach of Baker et al. (2016), but the object is broader economic risk rather than policy uncertainty. Article scores are graded instead of binary. A single article may enter more than one economic domain. ARS is built separately from AERI using bounded inputs for intensity, breadth, severity and active domains. The same scoring rule is applied across 54 economies with very different media depth and economic structures, and coverage is reported explicitly.

The paper also relates to work in applied econometrics that treats news text as a source of macroeconomic information (Ellingsen et al., 2022; Kalamara et al., 2022; Thorsrud, 2020). Those studies draw on deep archives from a single advanced economy. We take the same measurement problem to 54 economies whose archives differ in depth, language and continuity, and where the official series that text measures are usually checked against are often thin or late.

2 Literature Review

2.1 Economic Risk, Expectations, and Ex Ante Behaviour

Economic decisions depend on prospective outcomes as well as current conditions. Bloom (2009) shows how uncertainty can delay investment and hiring, while Jurado et al. (2015) measures uncertainty through the unpredictability of economic outcomes. These contributions motivate attention to information that appears before realised activity. A news-based risk index captures a different object: the salience and severity of economic concerns in reporting.

A clearly anticipated increase in financing costs may generate substantial risk reporting even when there is little uncertainty about whether it will occur. Conversely, a highly uncertain development may attract little coverage in a thin archive. AERI is built to capture the first of these margins: the amount and intensity of economic-risk content in the observed news flow. The empirical analysis then asks whether movements in that content bear systematic relationships to realised macroeconomic conditions.

2.2 Media, Information Frictions, and Macroeconomic Beliefs

Information reaches households and firms unevenly. Mankiw and Reis (2002) show how slow information diffusion can generate persistent inflation dynamics. Shiller (2017) emphasises narratives as a mechanism through which economic interpretations spread. News records changing attention and beliefs, but it also responds to realised conditions. The empirical analysis therefore examines movements from news to macroeconomic outcomes and feedback from realised conditions to the news measures.

An inflation response after inflation-risk reporting can arise for different reasons. News

may record cost pressures before their full price effects, reflect expectations about those pressures, or respond to information also used by firms. In the other direction, a rise in AERI after depreciation shows realised stress entering the news environment. The empirical design examines both relationships.

2.3 Newspaper-based indices of uncertainty and risk

Baker et al. (2016) construct economic policy uncertainty from articles jointly referring to the economy, policy and uncertainty. Caldara and Iacoviello (2022) develop a geopolitical-risk measure, and Ahir et al. (2022) use uncertainty-related language in country reports to extend international coverage. AERI retains the text-based approach but separates economic reporting into eight domains and incorporates severity and structural breadth. The domain structure shows whether reported stress is concentrated in particular parts of the economy rather than only whether a specified class of uncertainty terms appears.

For a text-based measure, the counted event and its denominator have to match the economic interpretation. AERI uses a graded article score, whereas ARS combines bounded information from several features of the monthly news flow. The eight pillars show where reported risk is concentrated. External-financing stress and domestic price pressure, for example, can rise at the same time without representing the same economic problem.

AERI also differs substantively from economic policy uncertainty. Its dictionary spans prices, production, finance and other economic domains, including developments whose occurrence may be relatively certain, and its article scores vary with severity and textual emphasis. EPU, by contrast, is built around joint references to the economy, policy and uncertainty. The two approaches share a news-based measurement tradition but represent different constructs and scales. Here, the broader economic-risk measure is implemented consistently across an African panel.

2.4 News text as an econometric measurement problem

A second body of work, much of it in applied econometrics and forecasting, treats newspaper text as data to be modelled and tested rather than only counted. Thorsrud (2020) extracts topics from a Norwegian business daily and combines them with quarterly GDP in a time-varying dynamic factor model. The resulting daily coincident index dates business-cycle phases closely and nowcasts output about as well as forecast combinations and expert judgement. Larsen and Thorsrud (2019) use the same archive to show that news topics predict a range of macroeconomic and financial variables, and that an index built from the most informative topics is followed by persistent movements in financial markets, credit and borrowing. Ellingsen et al. (2022) run a real-time forecasting comparison between 22.5 million Dow Jones articles and the FRED-MD database for US output,

consumption and investment growth. The news data carry information that the hard indicators miss, most clearly for consumption.

The choice of text method has received as much attention as the choice of data. Kalamara et al. (2022) compare dictionary counts with machine-learning methods applied to three UK newspapers. Text improves forecasts of GDP, inflation and unemployment, the gains are largest in periods of economic stress, and the best results come from term counts combined with supervised learning. Barbaglia et al. (2023) score only the text that is grammatically linked to an economic term of interest, using a lexicon written for economics and finance, and find that the resulting sentiment improves forecasts of US macroeconomic variables, including the tails of the forecast distribution. Shapiro et al. (2022) report that a lexicon designed for economic news outperforms general-purpose sentiment lists. Bybee et al. (2024) estimate a topic model on the full text of the *Wall Street Journal* and show that the attention paid to different themes closely tracks a wide range of economic activity. A related lesson runs through this work: word lists need to fit the domain. Loughran and McDonald (2011) show that a general-purpose negative-word list classifies many words as negative that are not negative in a financial context.

These papers share two features. Each draws on deep archives from a single advanced economy, and the more flexible methods learn their topics or weights from that archive or from a target series observed with reasonable accuracy. Neither feature carries over to our sample. Archive depth, language and outlet continuity differ widely between countries, and for many of them the official series that a supervised model would be trained on arrive late, are revised heavily or are themselves interpolated. A topic model estimated country by country would also produce topics that differ from one country to the next, so a high reading in one country would not carry the same meaning as a high reading in another. AERI therefore applies one published dictionary and one scoring rule, with English, French and Portuguese terms, across all 54 economies. The graded severity tiers and textual-emphasis weights bring the score closer to the intensity measures used in the sentiment literature while keeping each article's contribution traceable to explicit rules.

2.5 Composite indices, factor structure and missing observations

Composite measures of economic conditions have a long history in applied econometrics. Stock and Watson (2002) show that a small number of factors extracted from many series carry most of their forecasting content. Mariano and Murasawa (2003) and Aruoba et al. (2009) build coincident indices from indicators observed at different frequencies, and Lewis et al. (2022) construct a weekly index of US real activity from ten high-frequency series. Bańbura and Modugno (2014) extend maximum-likelihood factor estimation to data with an arbitrary pattern of missing observations, which is the usual situation when indicators start at different dates or are published with different delays.

Our aggregation step sits within this tradition but makes different choices for a different purpose. Gentzkow et al. (2019) treat textual measurement as an empirical design problem in which preprocessing, feature definition and interpretation affect the resulting economic variable. Dictionary-based scoring allows the contribution of an article to be traced to explicit rules. PCA then reduces the dimensionality of the aggregate inputs (Jolliffe & Cadima, 2016). It determines weights from covariance, rather than choosing them to maximise correlation with inflation or output, so the composite is not fitted to the outcomes it is later compared with. Economic interpretation continues to depend on the definitions of the inputs.

PCA enters only at the aggregation stage. Economic relevance is defined beforehand through the article-scoring rules. Standardisation prevents the numerical scale of an input from determining its principal-component loading, and converting the loadings back to the bounded input scale explains why a component with a smaller standardised loading can receive a larger final weight. We report the loadings, scaling bounds and final weights so that the empirical weighting step can be reproduced directly. Missing source months are kept as missing rather than filled in by a model. A factor model in the spirit of Bańbura and Modugno (2014) could estimate those months, but the published index would then depend on the model used to estimate them. Keeping the gaps visible leaves that choice to the user and keeps a missing archive month distinct from an observed month with no measured risk.

2.6 Cross-country measurement challenges and African contexts

Cross-country indices are sensitive to publication volume, source composition, language and archive access. Scaling by total article counts removes the direct level effect of media volume, while coverage diagnostics track variation in the underlying observation base. We also distinguish missing source months from observed months with zero measured risk. These choices are especially important when one statistical procedure is applied across heterogeneous media environments.

Comparability has both a time-series and a cross-sectional dimension. Within a country, changes in the available outlet set can affect the observed news mix; across countries, similar scores can arise from media systems with different degrees of economic-news specialisation. The common scoring rule provides definitional consistency and the minimum-coverage restriction limits the influence of very small denominators. The secondary empirical exercises further absorb persistent country differences and shared calendar shocks. Time-varying changes in national source coverage remain a separate measurement concern and are examined through the coverage diagnostics.

2.7 African Economies and Alternative Sources of High-Frequency Information

A growing literature uses alternative data to supplement conventional economic statistics where coverage is sparse or publication lags are long. Yeh et al. (2020) use publicly available satellite imagery and deep learning to measure local economic well-being across Africa. Buell et al. (2021) combine high-frequency indicators, including search and payments data, with statistical models to track economic activity in Sub-Saharan Africa. More recently, Fotopoulou et al. (2026) examine machine-learning and satellite-data methods for nowcasting real GDP in data-constrained settings. These studies illustrate a broader measurement problem: economically relevant information often arrives at frequencies and in forms that are not captured by standard national accounts releases.

News data capture another part of the high-frequency measurement problem by recording economic concerns as they appear in public reporting. The text-based forecasting studies reviewed above use archives from the United States, the United Kingdom and Norway, where conventional data are already timely. The case for news-based measurement is stronger where official statistics are thinner, and that is also where the measurement problems are hardest. The pillar indices keep price pressures, fiscal concerns, external financing conditions and the other domains separate, including months when the aggregate index changes little. We compare the news measures with official macroeconomic statistics and examine whether the timing and sign of the relationships match the definitions of the measures.

2.8 External Identification and Global Monetary Shocks

High-frequency monetary-policy identification lets us test AERI against a shock measured outside the news-scoring system. Jarociński and Karadi (2020) separate pure monetary-policy surprises from central-bank information shocks using the high-frequency co-movement of interest rates and equity prices around policy announcements. We interact their pure monetary-policy component with predetermined country exposure. The test asks whether the same global tightening produces a larger AERI response in countries that entered the sample with greater external financial exposure.

Inference in this design has to respect its structure. The shock varies over months and the exposure varies across countries, so errors may be correlated in both dimensions; two-way clustering by country and month allows for both (Cameron et al., 2011). The number of country clusters is small, however, and MacKinnon and Webb (2017) show that conventional cluster-robust t-tests can over-reject badly when clusters are few or differ widely in size, while the restricted wild cluster bootstrap generally performs better. We therefore read the exposure coefficient against finite-cluster reference distributions and report restricted wild-cluster-bootstrap results alongside them.

3 Measuring AERI and ARS

We build the measures from the collected articles, score each article, and then aggregate the scores by country and period. The output consists of AERI, eight overlapping pillar indices, an eight-pillar PCA common-factor composite, and the bounded ARS. Table 1 gives the definition and role of each measure.

Table 1: Measurement objects and their roles

Measure	Construction	Interpretation
AERI	100 times the sum of article-level risk scores divided by all retained articles in a country-month	Overall intensity of scored economic-risk content relative to the observed news flow; the principal economic-risk index
Pillar index	Same article-level risk score as AERI, with the numerator restricted to articles assigned to one of eight economic domains	Domain-specific diagnostic showing where measured economic risk is occurring; overlapping assignments preserve cross-domain connections
Eight-pillar PCA composite	Log-transformed, percentile-scaled pillar indices combined with fixed first-component weights	Common-factor diagnostic showing how strongly the eight risk domains move together; useful for distinguishing broad cross-domain stress from a concentrated pillar episode
ARS	Weighted average of bounded AERI intensity, relevant-article breadth, conditional severity and structural spread, scaled to 0–100	Multidimensional monitoring score summarising how intense, widespread, severe and broad-based the measured risk is

Notes: AERI is the principal article-score index. The pillar indices locate the economic domains in which that risk appears. The PCA composite provides a companion diagnostic of common cross-domain movement, while ARS is a separate bounded monitoring score. Each measure is interpreted according to its stated construction and role in the framework.

3.1 Data Collection

We assemble monthly country-level records from national news sources and organise retrieval by country, site and month, with bounded retrieval attempts and site-level diagnostics. The recorded retrieval settings allow the collection process to be repeated, although outlet coverage can still vary and repeated stories may remain. Corpus size is reported in retained article records. Section 4.5.2 evaluates the sensitivity of AERI and ARS to repeated identical article bodies using a separately reconstructed, internally matched archive.

Our analytical vintage contains 21,303,573 article records across 54 African countries from January 2015 to December 2025. Of these, 5,896,776 meet the economic relevance criterion. The 7,127 country-month records include 639 months with fewer than 100 articles. To limit sensitivity to very small news denominators, the principal descriptive comparisons use the remaining 6,488 country-months with at least 100 retained articles.[1] The median country-month contains 1,039 articles. Months for which the source archive is absent or unreadable are recorded as missing.

The country-month is the unit of aggregation. A widely syndicated event can therefore generate several article records, sometimes across multiple outlets. We keep record counts distinct from independent stories and apply the coverage threshold at the country-month level before making cross-country descriptive comparisons. Calibration, descriptive analysis and macroeconomic estimation use separately documented samples because their data requirements differ. Section 4.5.2 quantifies how much the published series changes when identical article bodies are collapsed within country-months.

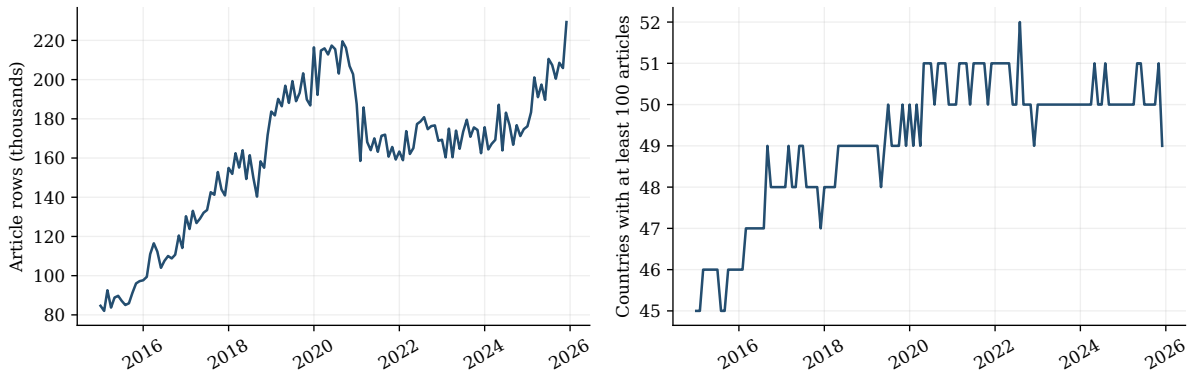


Figure 1: News coverage and eligible country observations

Notes: The left panel reports total retained article records by month. The right panel counts countries with at least 100 articles. Article counts refer to retained records before story-level exact-text deduplication. Changes in the archive can affect measured coverage.

3.2 Data Pre-processing

Preprocessing converts article text into economic relevance, severity, textual emphasis and pillar assignments. The same rules apply across the retained corpus. An article can activate several pillars because economic developments often span more than one domain.

3.2.1 Text Normalisation

The headline and body form a single textual unit, x_i. Normalisation removes formatting differences before dictionary matching. Scoring is rule-based and uses observable textual

[1] Months below the threshold remain in the underlying series and coverage files. The threshold governs the principal cross-country descriptive comparisons and the ARS calibration sample, while preserving the lower-volume observations in the documented archive.

features from this normalised text.

3.2.2 Economic Relevance Filtering

Relevance is defined by

$$R_i = \mathbf{1}\{x_i \text{ matches the economic-relevance dictionary}\}. \tag{1}$$

Non-relevant articles receive zero economic-risk score but remain in the denominator. Relevant reporting also includes discussions of economic conditions expressed without acute-crisis language.

All retained articles enter the denominator, so the share of the news flow devoted to economic risk remains part of the signal. The separate breadth and severity diagnostics then recover, respectively, the prevalence of relevant reporting and its conditional intensity. For an observed source month with retained records, a zero score is therefore an empirical outcome; missing source months are handled separately.

3.2.3 Severity Tiering

Let $\mathcal{K}_i \subseteq \{1, 2, 3, 4\}$ contain the severity tiers matched in the text. We set

$$S_i = \begin{cases} 0, & R_i = 0, \\ \max(\{1\} \cup \mathcal{K}_i), & R_i = 1. \end{cases} \tag{2}$$

A relevant article receives a baseline of one, with stronger matched language increasing the score. Where several cues occur, the highest tier applies. The resulting severity tier is an ordinal description of language strength.

3.2.4 Textual Emphasis Weighting

The textual weight combines a headline cue T_i, a body-length indicator L_i for at least 1,200 normalised characters, and an indicator M_i for at least two distinct active pillars:

$$W_i = \text{clip}(1 + 0.5T_i + 0.2L_i + 0.3M_i, 1, 2). \tag{3}$$

Here $\text{clip}(x, a, b) = \min(\max(x, a), b)$. The weight is a bounded textual-emphasis adjustment based on placement, length and cross-domain content. Its coefficients belong to the article-scoring rule and are distinct from the PCA aggregation weights used later in the construction.

3.2.5 Article-Level Risk Scoring

The score is

$$A_i = R_i S_i W_i, \qquad 0 \leq A_i \leq 8. \tag{4}$$

Each activated pillar contributes once to an article score; repeated phrases within a story therefore leave that pillar's contribution unchanged. Table 2 shows representative terms from the production dictionaries and a worked example of how the matching rules map one article into AERI, pillar and ARS inputs.

Table 2: Illustrative keyword taxonomy used in article scoring

Scoring layer	Illustrative terms from the production dictionary	Role in the computation
Economic relevance	economy, GDP, inflation, budget, debt, central bank, currency, bank, unemployment; French examples include *économie*, *PIB*, *dette*, *taux de change*; Portuguese examples include *economia*, *PIB*, *dívida*, *câmbio*	At least one match sets $R_i = 1$. Non-matching articles receive a zero article score but remain in the country-month denominator.
Risk pillars	Inflation: inflation, cost of living, shortage; FX/external: exchange rate, currency, devaluation, reserves; debt/fiscal: debt, default, restructuring, budget deficit; growth: recession, slowdown, unemployment; analogous dictionaries define the other four pillars	Pillar matches identify the domain composition of a relevant article. Pillars are non-exclusive and repeated mentions do not increase the pillar weight.
Severity tiers	Tier 1: concern, risk, uncertainty; Tier 2: surge, spike, worsening, deterioration; Tier 3: crisis, collapse, emergency; Tier 4: default, hyperinflation, bank run, economic collapse	A relevant article receives baseline severity 1; when several severity cues occur, S_i equals the highest matched tier.
Title/emphasis cues	crisis, default, devaluation, inflation, shortage, collapse, debt, restructuring, recession, bank, emergency	A title match adds 0.5 to W_i; a body of at least 1,200 normalised characters adds 0.2; two or more active pillars add 0.3. The weight is clipped to $[1, 2]$.

Notes: The table reports representative terms rather than the full dictionary. The supplied production configuration contains explicit English, French and Portuguese terms. Terms are assigned to relevance, pillar, severity and title-hint dictionaries before aggregation, independently of the subsequent macroeconomic validation outcomes.

Worked scoring example. Consider a hypothetical article titled "Inflation surge after currency devaluation." Suppose the normalised body exceeds 1,200 characters and contains no severity cue above tier 2. The article matches the economic-relevance dictionary, the

inflation and FX/external pillars, the tier-2 term "surge," and a title-risk cue. Hence $R_i = 1$, $S_i = 2$, $T_i = L_i = M_i = 1$, so

$$W_i = \text{clip}(1 + 0.5 + 0.2 + 0.3, 1, 2) = 2, \qquad A_i = R_i S_i W_i = 4.$$

If a country-month contains 100 retained articles and this is the only relevant article, this one record contributes $100(4/100) = 4$ points to AERI. The inflation and FX/external pillar indices each receive 4 points because the article contributes once to each active pillar. The same hypothetical month has breadth $B = 1/100 = 0.01$, conditional severity $V/4 = 2/4 = 0.50$, and structural spread $K/8 = 2/8 = 0.25$. With the fixed PCA ARS weights $(0.2625, 0.2722, 0.3223, 0.1430)$ and intensity $\text{AERI}/200 = 0.02$, the corresponding illustrative score is

$$\text{ARS} = 100\{0.2625(0.02) + 0.2722(0.01) + 0.3223(0.50) + 0.1430(0.25)\} = 20.49.$$

The example is purely mechanical and illustrates article-level scoring; monthly scores are formed from the full retained country-month article set.

3.3 Construction of the African Economic Risk Index

For country c and month t, let $\mathcal{A}_{ct}$ denote the retained articles and $N_{ct} = |\mathcal{A}_{ct}|$. AERI is

$$\text{AERI}_{ct} = \frac{100}{N_{ct}} \sum_{i \in \mathcal{A}_{ct}} A_i. \tag{5}$$

The index measures scored economic content per article. It can exceed 100, with a theoretical upper bound of 800. Annual AERI weights monthly values by article counts, so the annual numerator and denominator correspond to the same article universe.

Define $\overline{SW}^{R}_{ct} = \sum_i R_i S_i W_i / \sum_i R_i$ whenever relevant articles are present. AERI can then be written as

$$\text{AERI}_{ct} = 100 B_{ct} \overline{SW}^{R}_{ct}. \tag{6}$$

If there are no relevant articles, AERI is zero for a non-empty month. Equation (6) shows why rising AERI can reflect a larger share of economic-risk reporting, more severe or emphatic relevant reporting, or both. The decomposition also shows that breadth and intensity are related inputs to ARS. PCA summarises their covariance rather than treating the dimensions as statistically independent.

3.4 Pillar-Level Decomposition and Diagnostics

Let I_{pi} indicate that article i activates pillar p. The pillar index is

$$P_{pct} = \frac{100}{N_{ct}} \sum_{i \in \mathcal{A}_{ct}} A_i I_{pi}. \tag{7}$$

The eight pillars can overlap.[2] Each pillar lies between zero and AERI, and the sum can exceed AERI when a story contributes to several domains. Breadth, severity and structural coverage are

$$B_{ct} = \frac{\sum_i R_i}{N_{ct}}, \quad V_{ct} = \frac{\sum_i S_i}{\max(1, \sum_i R_i)}, \quad K_{ct} = \sum_{p=1}^{8} \mathbf{1}\{P_{pct} > 0\}. \tag{8}$$

Each raw pillar index measures one domain. To summarise their common cross-domain movement, define $\ell_{pct} = \log(1 + P_{pct})$ and

$$Q_{pct} = \text{clip}\left(\frac{\ell_{pct} - a_p}{b_p - a_p}, 0, 1\right), \quad C_{pct} = 100 u_p Q_{pct}, \quad H_{ct} = \sum_{p=1}^{8} C_{pct}. \tag{9}$$

The bounds a_p and b_p are the first and ninety-ninth calibration percentiles, and u_p are fixed weights obtained from the first principal component of the standardised transformed pillars. The resulting H_{ct} is the eight-pillar PCA composite, which we use as a common-factor diagnostic. AERI can be high when overall scored risk content is elevated but concentrated in a few domains; the PCA composite rises most strongly when several transformed pillar indices move together. The comparison helps distinguish broad cross-domain stress from episodes driven mainly by one or two pillars.

The PCA composite captures cross-domain co-movement that $K_{ct}/8$ does not. The ARS component $K_{ct}/8$ counts active pillars; H_{ct} also incorporates their transformed intensity and empirical common-factor weights. Table 4 reports the corresponding loadings, scaling bounds and weights.

3.5 Construction of the African Risk Score

ARS combines four bounded inputs:

$$X_{ct} = \left(\text{clip}\left(\frac{\text{AERI}_{ct}}{200}, 0, 1\right), B_{ct}, \frac{V_{ct}}{4}, \frac{K_{ct}}{8}\right)', \qquad \text{ARS}_{ct} = 100 \sum_{j=1}^{4} w_j X_{jct}. \tag{10}$$

[2]This treatment parallels the category-specific newspaper indexes in Baker et al. (2016), where a single article may enter more than one category.

The divisor 200 sets the saturation point for the intensity component. The four entries have direct interpretations: $\text{clip}(\text{AERI}/200, 0, 1)$ is bounded AERI intensity, B is the share of retained articles that are economically relevant, $V/4$ scales conditional severity to the unit interval, and $K/8$ records the share of the eight domains that are active. ARS is therefore a bounded 0–100 summary of the reported risk environment, rising as risk-related reporting becomes more intense, prevalent, severe or structurally widespread.

The four components are allowed to covary. Breadth is already one margin of AERI through equation (6), but it enters ARS separately by design. This allows two country-months with similar AERI to receive different ARS values when one reflects narrow, highly intense reporting and the other reflects broader, multi-domain stress. PCA summarises the covariance among these diagnostics and reduces the dependence of the final score on arbitrary hand-set weights.

Standardise the calibration inputs as $Z_j = (X_j - \mu_j)/s_j$, using population standard deviations. Let v be the unit-length first principal-component loading vector, oriented toward higher component values. The weights on the bounded input scale are

$$v = \arg\max_{\|b\|=1} b'\widehat{\Sigma}_Z b, \qquad w_j = \frac{v_j/s_j}{\sum_k v_k/s_k}. \tag{11}$$

Dividing by s_j converts standardised loadings to coefficients on the bounded input scale. With positive loadings, ARS is a rescaled first component and is weakly increasing in its bounded inputs. The same conversion is applied separately to the transformed pillar vector Q to obtain the fixed common-factor weights u_p used in equation (9). Calibration rejects near-constant inputs or materially mixed-sign loadings rather than converting negative loadings to absolute values.

We calibrate ARS using 6,238 country-months with at least 100 articles during January 2015–July 2025. ARS weights are 26.25 percent for intensity, 27.22 percent for breadth, 32.23 percent for severity and 14.30 percent for structure. PC1 explains 62.22 percent of standardised variation in the four ARS inputs and 52.50 percent for the transformed eight-pillar system. Weights remain fixed between recalibrations. Historical scores remain retrospective because calibration uses information through July 2025. In the eligible 2015–2025 sample, the median month activates seven of the eight pillars, while the median relevant-article share is 25.8 percent. These diagnostics help explain why ARS can move differently from AERI even though both are generated from the same underlying article corpus.

Tables 3 and 4 report the two PCA calibrations. Severity receives the largest ARS weight even though intensity has the largest standardised loading because equation (11) converts standardised loadings back to coefficients on the bounded input scale before normalisation. Table 4 documents how the transformed domain indices load on their common factor and makes the construction used in Figure 3 auditable. Debt and fiscal

risk and financial-sector risk receive the largest common-factor weights, at about 17.3 percent each, while commodity and supply risk receives 6.48 percent.

Calibration pools eligible country-month rows without imposing equal total country weights. Countries with more eligible months consequently contribute more observations. Holding the calibration fixed makes subsequent changes in a score attributable to changed inputs rather than to month-by-month re-estimation of its weights. A future recalibration should be treated as a new measurement vintage, with the associated parameters retained. Because ARS compresses several diagnostics into one score, the individual pillar and component series remain necessary for interpretation; information outside the common movement of the four ARS inputs may still be economically important.

Table 3: ARS component weights and standardised loadings

Component	Benchmark (%)	PCA (%)	PC1 loading
Intensity	40.00	26.25	0.5803
Breadth	25.00	27.22	0.5502
Severity	20.00	32.23	0.3921
Structure	15.00	14.30	0.4548

Notes: The 6,238 calibration observations cover January 2015–July 2025 and have at least 100 articles. PC1 explains 62.22 percent of standardised variation. Weights on the bounded input scale differ from standardised loadings; equation (11) gives the conversion.

Table 4: PCA weights, scaling bounds and loadings for the eight pillars

Pillar	Weight (%)	Log lower	Log upper	PC1 loading
Inflation	12.95	0.0000	2.9629	0.4052
FX and external	13.21	0.0000	4.2958	0.3584
Debt and fiscal	17.29	0.0000	4.3550	0.4071
Financial sector	17.30	0.0000	4.5177	0.4219
Growth and real economy	11.90	0.0000	2.6486	0.3861
Commodity and supply	6.48	0.0000	1.0248	0.1289
Political economy	9.92	0.0000	2.9873	0.3063
Climate and disaster	10.95	0.0000	2.3585	0.3227

Notes: Bounds are the first and ninety-ninth calibration percentiles of $\log(1 + P_p)$. PC1 explains 52.50 percent of standardised variation. The weights quantify each transformed pillar's contribution to the common cross-domain factor after scale adjustment; economic interpretation remains anchored in the domain-specific series and results. Calculations use unrounded weights, which sum to one.

4 Empirical Properties and Evaluation

4.1 Aggregate Risk Dynamics

Figure 2 presents the two principal measures used throughout the paper. The left panel shows AERI as defined in equation (5); the right panel shows ARS from equation (10), using the fixed PCA weights in Table 3. For each month, the figure plots the unweighted cross-country median with interquartile ranges as shaded bands. The eight-pillar PCA composite is analysed separately in Figure 3.

The monthly median AERI reaches its sample maximum of 68.94 in April 2020, coinciding with the acute phase of the COVID-19 pandemic and the associated economic disruption. The shaded bands show dispersion across eligible countries. Because the set of eligible countries changes with archive coverage, aggregate movements are read alongside Figure 1.

Across 6,488 eligible country-months, AERI has a mean of 47.89 and a standard deviation of 27.56. ARS has a mean of 36.59 and a standard deviation of 9.66. Their different scales follow directly from construction: AERI preserves article-score intensity, while ARS combines bounded components. The two measures are read jointly, with AERI describing the level of the scored signal and ARS its broader configuration.

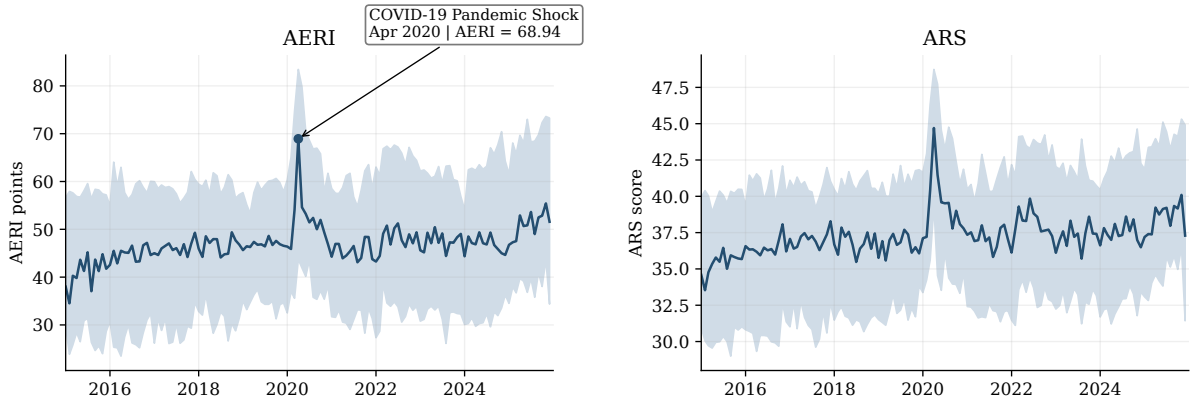


Figure 2: Aggregate AERI and ARS dynamics

Notes: Lines are unweighted country medians; shaded regions are interquartile ranges. The annotation marks the COVID-19 pandemic shock at the maximum monthly median AERI in the sample (April 2020, 68.94). Only country-months with at least 100 articles enter. Neither series is weighted by GDP. ARS uses the fixed PCA calibration.

Figure 3 makes two comparisons. Panel A compares AERI with the eight-pillar PCA composite, which summarises common movement across the transformed domain indices. Their monthly-median correlation is 0.558 over 2015–2025, and both reach their maximum in April 2020. The common peak indicates that the dominant AERI episode coincided with unusually broad movement across risk domains. Outside that episode the series diverge in informative ways. May 2022, for example, is the second-highest monthly median

of the pillar composite but falls outside the five highest AERI months, pointing to stronger cross-domain co-movement than aggregate article-score intensity alone would suggest.

Panel B isolates the weighting choice. The benchmark 40/25/20/15 score and the PCA-weighted score use the same inputs and observations; their monthly cross-country medians correlate 0.975 and both attain their 2015–2025 maximum in April 2020. The broad ARS chronology changes little under this substantially different weighting scheme.

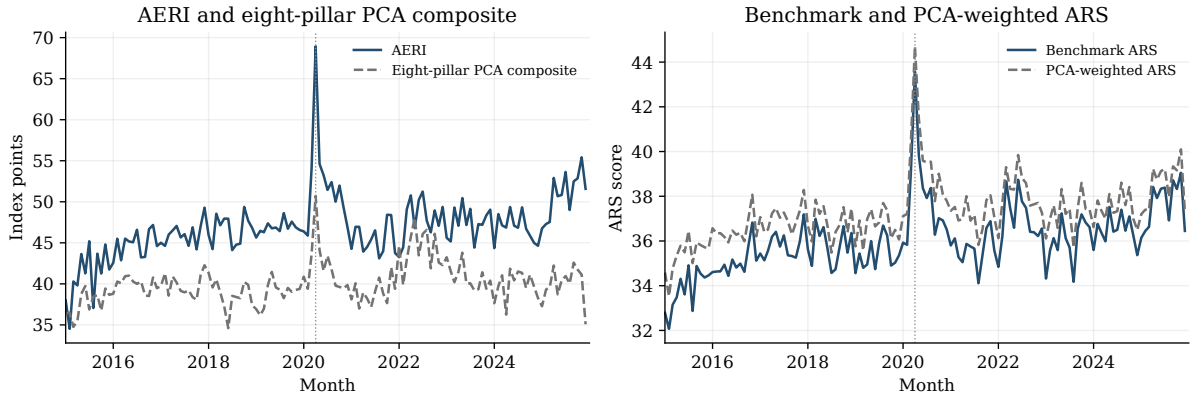


Figure 3: Weighting sensitivity of the AERI-related and ARS measures

Notes: Panel A compares AERI with the eight-pillar PCA common-factor composite. Panel B compares benchmark ARS, using 40/25/20/15 weights on intensity, breadth, severity and structure, with PCA-weighted ARS. Lines are monthly unweighted country medians for country-months with at least 100 retained articles. The vertical reference marks April 2020. The monthly-median correlations are 0.558 for AERI versus the eight-pillar PCA composite and 0.975 for benchmark versus PCA-weighted ARS.

Table 5: Distribution of AERI and alternative ARS specifications

Measure	Mean	SD	Median	Minimum	Maximum
AERI	47.89	27.56	46.91	0.00	300.16
ARS (PCA weights)	36.59	9.66	37.27	0.00	85.37
Benchmark ARS	35.42	10.71	36.01	0.00	88.93

Notes: Common sample: 6,488 country-months with at least 100 articles, 2015–2025. The benchmark and PCA-weighted ARS have Pearson correlation 0.9950, rank correlation 0.9971, and mean absolute difference 1.49 points. The comparison changes only the ARS weights; AERI itself has no PCA weighting stage.

Table 5 shows the same pattern as Figure 3. The PCA-weighted and benchmark ARS have very similar central tendencies and nearly identical rankings, although the PCA specification is modestly less dispersed (standard deviation 9.66 versus 10.71). Changing the weights shifts the scale slightly while leaving the relative ordering of country-months largely intact. AERI is considerably more dispersed because it preserves the original article-score intensity and remains on its own scale.

4.2 Annual Africa Economic Risk Indices by Country (2015–2025)

Figure 4 follows annual AERI for the ten economies examined in the country discussion. The common colour scale makes differences in intensity and persistence visible without imposing a common economic explanation on each episode. Annual values aggregate the monthly article-score numerator and denominator. Among countries with at least nine eligible months in 2025, the highest annual AERI values are Zambia (112.40), Liberia (99.59), Lesotho (91.14). Annual values are article-count-weighted averages of monthly AERI scores. The resulting ordering describes the intensity of observed economic-risk reporting across the retained news archive. We complement the annual comparison with exact monthly summary statistics for the same ten-country group in Appendix Table E.1; the full all-country summary and monthly country paths are reported in the accompanying Supplementary Material. These outputs document differences in levels, variability and the timing of peaks that the aggregate median and annual averages conceal. Within the ten-country group, Nigeria has the highest mean monthly AERI, while Ethiopia exhibits the greatest time-series dispersion. These country results are descriptive and use the same AERI definition as the aggregate analysis.

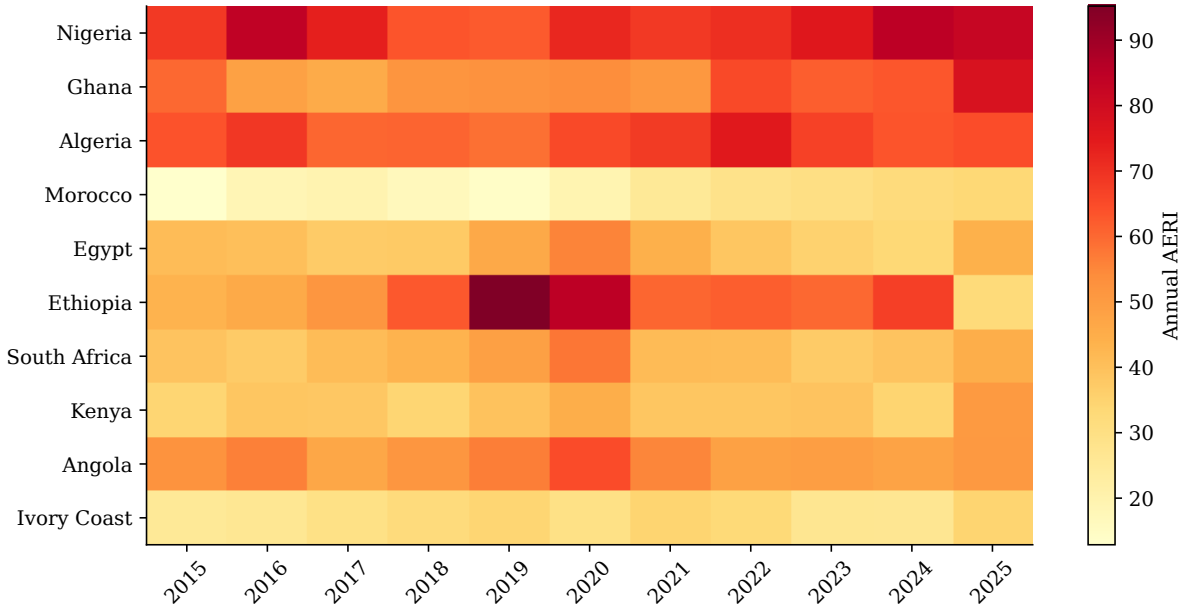


Figure 4: Annual AERI in selected African economies

Notes: The figure reports annual AERI for the ten illustrative economies over 2015–2025. Monthly AERI values are aggregated using monthly retained-article counts, so the annual statistic preserves the article-score numerator and denominator. The ten countries form the illustrative country group used in the descriptive discussion, and the underlying annual file covers all 54 economies. The common colour scale highlights substantial cross-country differences in levels, persistence and the timing of peaks.

Figure 4 shows that the aggregate African pattern hides marked country differ-

ences. Angola, Egypt, Ethiopia, Kenya and South Africa all record higher annual AERI in 2020 than in their own pre-pandemic averages. Nigeria stays comparatively high through much of the sample and reaches its annual maximum in 2024. Morocco rises gradually after 2020 rather than showing a single pandemic-centred jump, while Ethiopia peaks in 2019 and remains elevated in 2020. The continental median captures the common timing of some episodes, but it averages over persistence and country-specific peaks.

4.3 Pillar-Level Decomposition

The pillar indices decompose the aggregate signal by economic domain. Figure 5 reports the pooled mean of each raw pillar index across the 6,488 eligible country-months, and Table 6 shows how the domain profile changes across broad subperiods. Because an article may activate several domains, these are overlapping measures of reported risk content rather than shares of a fixed total.

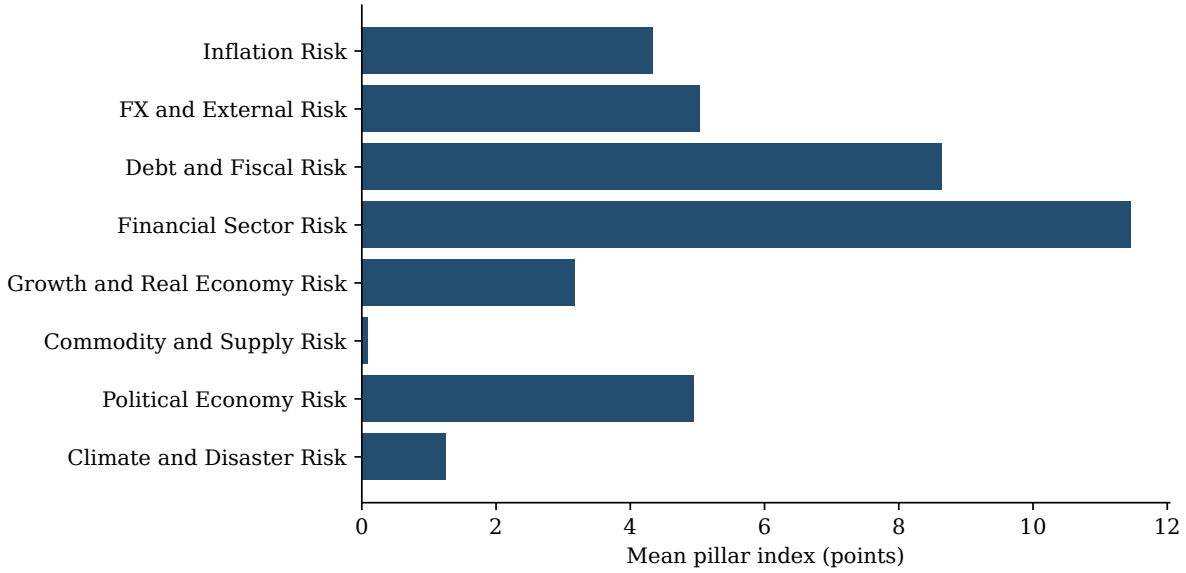


Figure 5: Average domain-specific pillar indices

Notes: Bars are ordered by magnitude and report pooled means of the eight raw pillar indices across 6,488 country-months with at least 100 retained articles. Each pillar uses the same graded article score as AERI, with the numerator restricted to articles assigned to that domain. Pillars are non-exclusive, so a single article may contribute to several domain indices. Financial-sector and debt/fiscal reporting have the largest pooled means in this vintage.

Financial-sector risk has the largest pooled mean, followed by debt and fiscal risk. FX/external, political-economy and inflation risks form a middle group; growth and real-economy risk sits below them, while climate/disaster and commodity/supply risks are the least prominent on average. Dictionary size does not account for the financial-sector ranking: that pillar contains nine top-level terms or phrases, the same number as six other pillars, while commodity/supply contains seven. Term frequency, outlet specialisation

and the newsworthiness of particular events can differ across domains. For that reason, the level comparisons are descriptive, and the within-pillar changes over time carry more weight in interpretation.

Table 6: Pillar-specific economic-risk content over time

Pillar	2015–2019	2020	2021–2025	Overall
Inflation risk	3.47	3.49	5.33	4.34
FX and external risk	5.59	5.11	4.48	5.03
Debt and fiscal risk	9.47	9.64	7.65	8.64
Financial-sector risk	11.75	12.87	10.90	11.46
Growth and real economy risk	3.14	4.85	2.85	3.17
Commodity and supply risk	0.07	0.03	0.13	0.09
Political-economy risk	5.27	5.51	4.50	4.94
Climate and disaster risk	1.25	1.11	1.27	1.25

Notes: Entries are means of the raw pillar indices on the same score-per-article scale used in AERI, computed over eligible country-months. The pillars are non-exclusive: one article may contribute to more than one domain, and the rows describe overlapping domain-specific content. The period split is descriptive and highlights changes in the composition of reported economic risk.

The subperiod means show the same pattern. Financial-sector and debt/fiscal risk remain the two largest domains in each reported period, but the composition changes around major episodes. Growth and real-economy risk rises from 3.14 in 2015–2019 to 4.85 in 2020, while inflation risk rises from 3.47 before the pandemic to 5.33 in 2021–2025. FX/external and debt/fiscal risk are somewhat lower on average in 2021–2025 than before 2020. The pillar system distinguishes changes in the composition of risk that the aggregate AERI series alone would obscure.

4.4 From AERI to ARS: Intensity, Breadth, and Structure

ARS separates reporting intensity from the other components of the monthly news signal. Structural breadth is the number of domains with positive scores and can remain high even when AERI is low. Comparing the two measures shows whether an AERI increase is concentrated in a small number of domains or accompanies wider economic coverage.

A fixed-weight benchmark using 40/25/20/15 percent for intensity, breadth, severity and structure has a pooled Pearson correlation of 0.995 with PCA-weighted ARS and a rank correlation of 0.9971. Their mean absolute difference is 1.49 points, showing that the broad ARS pattern is highly insensitive to the weighting choice. The pooled correlation between AERI and PCA-weighted ARS is lower, at 0.911. They clearly share a common signal, but ARS changes the scale and gives explicit weight to breadth, severity and structural spread.

Table 5 provides the corresponding distributional comparison. We hold the sample and component definitions fixed, so the exercise isolates the effect of the weights them-

selves. The PCA specification supplies a reproducible empirical weighting rule, while the 40/25/20/15 benchmark provides a simple reference point. The two schemes diverge most when intensity and severity move differently, because those components receive the largest differences in weight. A country-month with high conditional severity but only moderate AERI intensity will tend to score higher under the PCA weights; an episode dominated by unusually high intensity with less exceptional severity will tend to score higher under the benchmark.

4.5 Evaluation of AERI and ARS

Following Baker et al. (2016), evaluation starts with the accounting identities and the coverage record. We then examine duplicate stories and alternative ARS weights, compare the series with salient economic episodes, estimate selected PVAR relationships, and use externally identified U.S. monetary-policy shocks interacted with predetermined country exposure. Each exercise addresses a different part of the measurement problem.

4.5.1 Accounting, coverage and aggregation robustness

We reconcile AERI directly to its article-score numerator and total-article denominator. Equation (6) separates the relevance-share margin from conditional severity and emphasis. Pillar scores use the same article score with overlapping domain indicators, and ARS combines four bounded diagnostics under a frozen calibration. The published country-month measures can be traced back to those article-level calculations.

Coverage also matters. The 2015–2025 vintage contains 7,127 country-month records, of which 6,488 meet the 100-article threshold used for the principal descriptive comparisons. Lower-volume months remain in the documented archive but are omitted from those comparisons because ratio-based measures become unstable with very small denominators. Figure 1 shows the resulting variation in the observation base. As an archive-depth diagnostic, we regress AERI on log retained article volume within the eligible sample. The within-country correlation is 0.035; the article-volume coefficient is 1.14 with country fixed effects ($p = 0.702$) and -1.47 after adding year-month fixed effects ($p = 0.640$). Appendix Table B.1 reports the specifications. The estimates show little association between AERI and archive size within countries. Changes in source composition and reporting practice remain separate concerns.

We assess ARS weight sensitivity by replacing the PCA weights with the benchmark 40/25/20/15 rule. The broad score pattern is almost unchanged: Pearson correlation is 0.9950, rank correlation is 0.9971 and the mean absolute difference is 1.49 points. With the sample and component definitions held fixed, the exercise attributes the remaining differences to the weighting choice.

4.5.2 Syndicated stories and exact-text deduplication

Repeated publication can amplify a country-month signal when the same high-scoring text appears in several outlets. We collapse identical decoded article bodies within each country-month, retaining one representative per duplicate group.[3]

The correction is applied algebraically to the frozen analytical vintage. Each removed record is scored with the original dictionaries, and its contribution is subtracted from the archived numerator, relevance and severity counts and each pillar numerator. The all-article denominator is reduced by the same record count. Source files are read-only, and duplicate counts, representative examples and file fingerprints are retained in the replication extension.

Across 21,303,020 frozen-vintage records represented in 7,126 reconstructed country-months, the rule removes 497,091 repetitions (2.33%), including 186,379 cross-outlet copies. The reconstruction is 553 records and one Togo country-month short of the archived aggregate because 42 supplied source files were empty or unreadable. For months with a partial count mismatch, unmatched frozen-vintage records remain in the denominator and only directly observed duplicates are removed. On the common coverage sample, original and duplicate-adjusted AERI have Pearson correlation 0.9296 and rank correlation 0.9883. The mean absolute change is 2.32 AERI points, and the 95th-percentile absolute change is 8.30. Fixed-weight ARS has correlation 0.9680 and mean absolute change 0.61 points. 69 country-months that originally met the 100-article rule fall below it after removal.

Table 7: Sensitivity to within-country-month exact-text deduplication

Statistic	Result
Records removed	497,091
Removed share (percent)	2.33
Cross-outlet copies removed	186,379
AERI Pearson correlation	0.9296
AERI rank correlation	0.9883
AERI mean absolute change	2.32
AERI 95th-percentile absolute change	8.30
ARS Pearson correlation	0.9680
ARS mean absolute change	0.61
Months falling below 100 articles	69

Notes: The comparison uses 6,418 country-months with at least 100 articles before and after deduplication. Exact duplicates have the same decoded body within a country-month; bodies shorter than 200 characters are retained. Both numerator and denominator are adjusted. ARS uses the original unrounded PCA weights.

Country-level results show where duplication matters most. The median within-country AERI correlation is 0.9992, and 48 of 53 countries retain the same peak month on the

[3]Exact matching is applied to decoded bodies of at least 200 characters. Appearances in different countries or months remain separate news exposures, while minor edits, translations and rewritten versions remain distinct records.

common sample. Sierra Leone's original November 2016 peak falls from 269.72 to 110.77 after removal; Rwanda's August 2021 peak falls from 95.43 to 63.35; and Mauritius's May 2020 peak rises from 91.64 to 111.79 because duplicate removal changes both the numerator and the denominator. At the cross-country level, the median still peaks in April 2020. The broad chronology changes little after deduplication, although a small number of country-specific spikes are sensitive to identical reprints. The PVAR and external-shock specifications continue to use the original benchmark vintage.

4.5.3 Event correspondence

The aggregate AERI series reaches its sample maximum in April 2020, when the monthly cross-country median is 68.94. The timing coincides with the acute phase of the COVID-19 pandemic and provides a simple check on the economic content of the index. Country peaks occur in different months, however. Reading the aggregate series together with country paths and pillar composition reveals whether a spike reflects a broad continental episode or a more localised event.

4.5.4 PVAR-based external-consistency evidence

We use pooled PVARs and impulse-response functions to compare the news measures with subsequent macroeconomic outcomes. Inflation-risk reporting is paired with realised inflation, exchange-rate innovations with AERI, and monthly GDP growth with AERI. Table 8 reports these relationships. All three joint lag tests are statistically significant at the paper's 10 percent criterion, with the inflation-risk and depreciation relationships also significant at 5 percent. At the reported horizons, the pointwise 90 percent intervals exclude zero in the predicted direction.[4]

Table 8: Directional PVAR external-consistency tests

Relationship	Expected	Lag-test p	Key response (90% interval)
Inflation risk → headline inflation	+	0.0299	M6: +0.531 pp [0.287, 0.825]
Depreciation → AERI	+	0.0142	M2: +0.0166 SD [0.0060, 0.0241]
GDP growth → AERI	−	0.0624	M1: −0.181 SD [−0.338, −0.068]

Notes: Expected signs are implied by the economic interpretation of the measures rather than chosen from the estimated responses. All three joint lag tests are statistically significant at the paper's 10 percent criterion, and the reported pointwise 90 percent intervals exclude zero in the expected direction. The inflation-risk and depreciation lag tests are also significant at 5 percent. Exact p-values are shown so readers can apply alternative significance conventions. These are conditional dynamic associations under the stated recursive ordering; structural identification is examined separately with the external monetary-policy shock design.

[4]The bootstrap intervals are pointwise across horizons. They quantify horizon-specific uncertainty; simultaneous coverage of an entire impulse-response path would require a different band construction.

The strongest forward-looking result concerns the inflation-risk pillar. In the GDP-augmented PVAR, the three inflation-risk lags are jointly significant in the headline-inflation equation ($p = 0.0299$). A one-within-country-standard-deviation inflation-risk innovation raises headline inflation by 0.111 percentage points on impact, 0.288 after one month and 0.392 after two months. The response continues to build and reaches 0.531 percentage points in month six, with a 90 percent interval of $[0.287, 0.825]$. The pointwise 90 percent band stays above zero at every horizon from impact through month twelve. The hump-shaped profile indicates that the inflation-risk pillar contains information about sustained price pressure beyond the contemporaneous inflation reading. In the terminology of Baker et al. (2016), the news-based innovation *foreshadows* subsequent inflation.

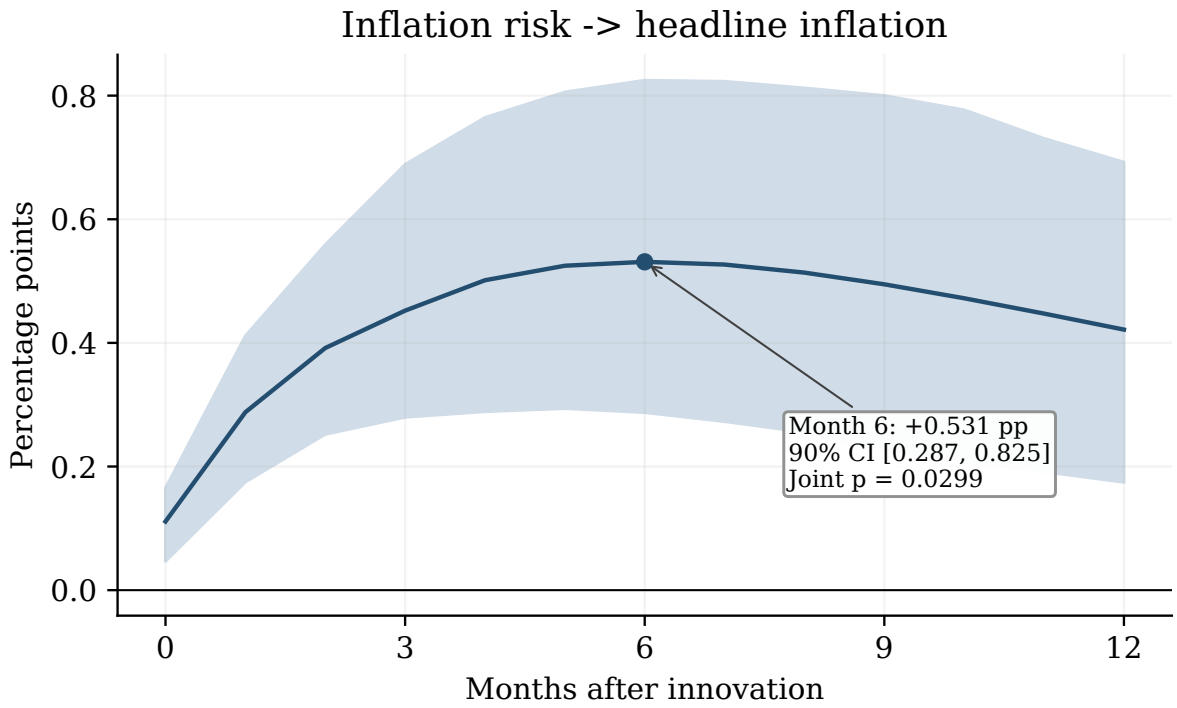


Figure 6: Headline inflation response to an inflation-risk innovation
Notes: The figure uses the 36-country GDP-augmented PVAR(3), estimated on 4,286 observations. The innovation is a one-within-country-standard-deviation increase in the article-score inflation-risk pillar. Bands are pointwise 90 percent intervals from 400 country-cluster bootstrap draws. The three inflation-risk lags are jointly significant in the headline-inflation equation ($p = 0.0299$); the response is positive at every reported horizon and peaks at 0.531 percentage points in month 6.

Figure 7 examines feedback from realised macroeconomic conditions to AERI. The exchange-rate result is particularly clear. A one-percentage-point innovation in monthly depreciation raises AERI by 0.0098 within-country standard deviations on impact, with a 90 percent interval of $[0.0033, 0.0153]$. After a small month-one response, AERI rises to 0.0166 standard deviations in month two, with a 90 percent interval of $[0.0060, 0.0241]$. The three depreciation lags are jointly significant in the AERI equation ($p = 0.0142$). The response is short-lived rather than permanent. Its sign and timing match the interpretation

of AERI as a measure that can register currency stress as exchange-rate pressure enters the news.

The real-activity response provides a complementary sign test. A one-percentage-point innovation in monthly GDP growth has a small and imprecise impact response, but AERI falls sharply one month later by 0.181 within-country standard deviations, with a 90 percent interval of $[-0.338, -0.068]$. Point estimates remain negative thereafter and gradually return toward zero. The joint three-lag Wald test is $p = 0.0624$, which is statistically significant at the 10 percent level. The horizon-specific response and the joint lag test therefore point in the same direction: stronger activity is followed by less intense economic-risk reporting once information about improved real conditions is absorbed into the news flow.

Across the three IRFs, inflation-risk reporting precedes higher realised inflation, currency depreciation is followed by higher AERI, and stronger real activity is followed by lower AERI. The forward inflation system places the risk measure first; the feedback system places the macroeconomic variables first. The same sign pattern appears under both recursive orderings. We interpret the estimates as conditional dynamic relationships. Structural identification is handled separately in the external-shock exercise below.

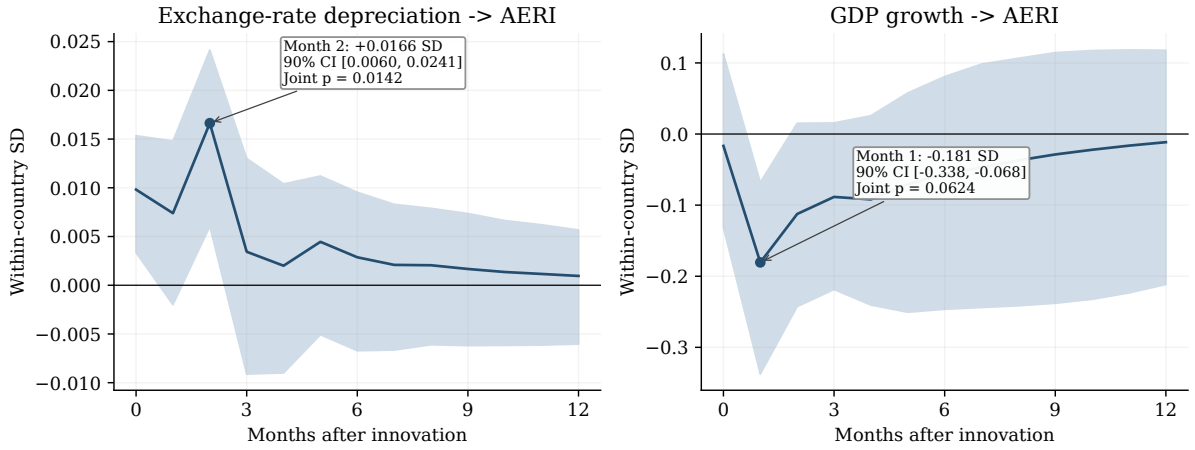


Figure 7: AERI responses to exchange-rate and real-activity innovations

Notes: Reverse 36-country GDP-augmented PVAR(3), estimated on 4,286 observations. Depreciation raises AERI, peaking at 0.0166 within-country standard deviations in month 2; GDP growth is followed by a −0.181 standard-deviation AERI response in month 1. Shading denotes pointwise 90 percent country-cluster bootstrap intervals. The opposite signs are consistent with higher reported risk under currency stress and lower reported risk after stronger activity.

4.5.5 Externally identified monetary-policy exposure test

To complement the recursive PVAR, we turn to a shock measured outside the AERI system. Following Jarociński and Karadi (2020), we use the pure U.S. monetary-policy surprise extracted from high-frequency movements in interest rates and stock prices around Federal Reserve announcements. We aggregate the surprise to the month and standardise it over 2015–2025. Country exposure is fixed at 2014 external debt stocks as a share of GNI from the World Bank International Debt Statistics (World Bank, 2026), so the exposure measure predates the AERI sample.

For country i and horizon h, the preferred local projection is

$$\begin{aligned}\text{AERI}_{i,t+h} - \text{AERI}_{i,t-1} = \alpha_i^h + \gamma_t^h + \beta_h \left(MP_t \times E_i^{2014}\right) \\ + \sum_{\ell=1}^{3} \theta_{h\ell} \Delta\text{AERI}_{i,t-\ell} + \sum_{\ell=1}^{3} \psi_{h\ell} \left(MP_{t-\ell} \times E_i^{2014}\right) + \varepsilon_{i,t+h}.\end{aligned} \tag{12}$$

where MP_t is a one-standard-deviation pure monetary-policy tightening and E_i^{2014} is standardised log(1 + external debt/GNI). Country fixed effects absorb persistent differences in average risk reporting, while year-month fixed effects absorb the common global shock, COVID-19 and other disturbances shared across countries in a given month. The identifying variation is therefore cross-country exposure to the same externally measured tightening. Standard errors are two-way clustered by country and month following Cameron et al. (2011). We use finite-cluster reference inference and report restricted wild-cluster-bootstrap results as an additional finite-sample check.[5]

Our primary sample contains 29 African economies with both AERI and 2014 external-debt exposure. Figure 8 shows a delayed differential response. Four months after a one-standard-deviation tightening, a country one standard deviation more exposed before the sample records a 0.272-point larger cumulative AERI increase. The finite-cluster 90 percent interval is $[0.108, 0.436]$ and $p = 0.0086$. By month nine the differential reaches 0.384 points, with interval $[0.085, 0.684]$ and $p = 0.0373$. Restricted wild-cluster-bootstrap p-values are 0.042 and 0.034 at months four and nine, respectively. The response is delayed rather than contemporaneous, a pattern consistent with a gradual adjustment of the reported risk environment following tighter external financing conditions.

Table 9 reports alternative exposure definitions, sample changes and a different monetary-policy shock. Every month-four and month-nine estimate remains positive and significant at the paper's 10 percent criterion under finite-cluster inference. The month-four coefficient also remains significant at 5 percent across all six reported variants. Month-nine estimates vary more in magnitude but retain a positive sign and 10 percent

[5]The identifying cross-section contains 29 country clusters, so the finite-cluster t and F references use 28 denominator degrees of freedom. The restricted wild-cluster bootstrap follows MacKinnon and Webb (2017).

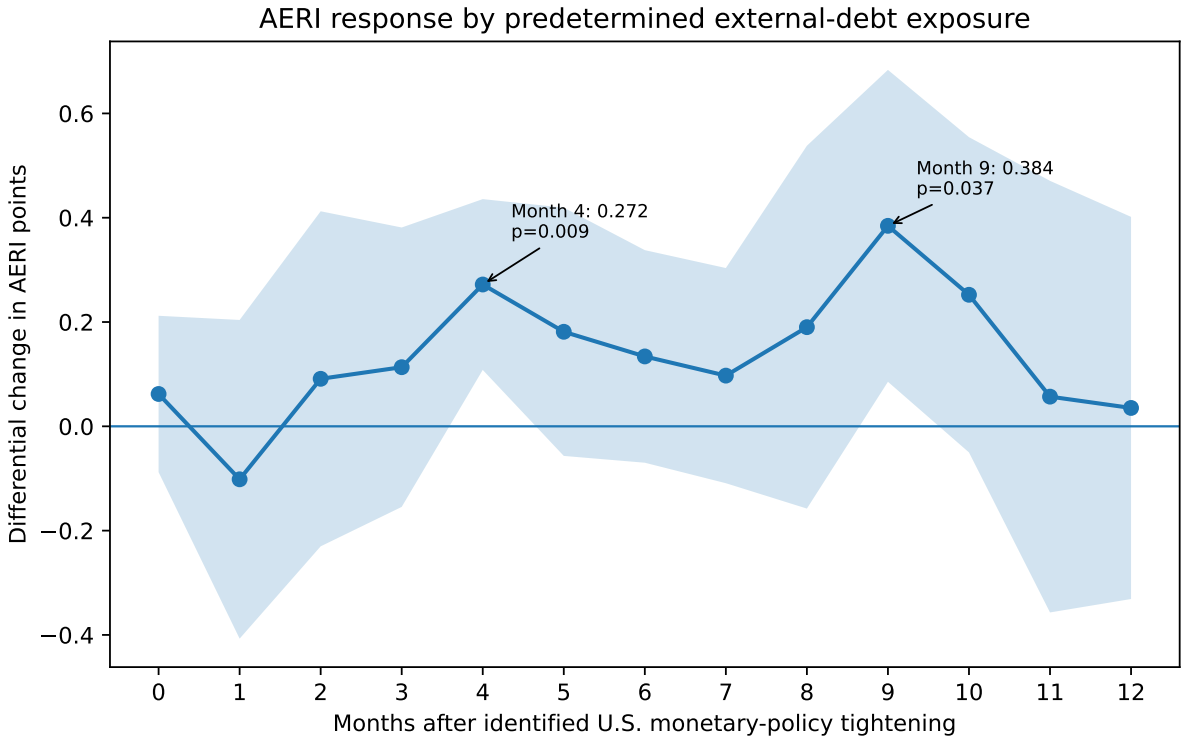


Figure 8: AERI response to an externally identified U.S. monetary-policy tightening by predetermined external-debt exposure

Notes: The sample contains 29 African economies with AERI and 2014 external-debt exposure. The line reports $\hat{\beta}_h$ from equation (12); shaded bands are pointwise 90 percent finite-cluster intervals using 28 denominator degrees of freedom. The shock is the standardised pure monetary-policy surprise from Jarociński and Karadi (2020), and exposure is standardised $\log(1 + 2014 \text{ external debt/GNI})$. The dependent variable is the cumulative change in AERI from month $t - 1$ through horizon h. Country and year-month fixed effects and three lags of the outcome and interaction are included; standard errors are two-way clustered by country and month. The differential response is positive and statistically significant at months 4 and 9 under the finite-cluster reference.

Table 9: Externally identified monetary-policy exposure validation

Specification	$\hat{\beta}_4$	p_4	$\hat{\beta}_9$	p_9
Primary: log debt/GNI	0.272	0.0086	0.384	0.0373
Raw debt/GNI	0.203	0.0152	0.289	0.0556
95th-percentile-capped debt/GNI	0.230	0.0284	0.347	0.0566
Exclude Mozambique	0.265	0.0305	0.390	0.0610
African-country panel restricted to FOMC announcement months	0.299	0.0078	0.354	0.0395
Alternative monetary-policy shock	0.251	0.0068	0.420	0.0534
Leave-one-country-out range	[0.214, 0.308]	29/29 $p < 0.10$	[0.282, 0.455]	29/29 $p < 0.10$

Notes: The dependent variable is the cumulative change in AERI from month $t - 1$ through horizon h. The shock is the standardized pure U.S. monetary-policy surprise from the Jarociński–Karadi high-frequency decomposition, aggregated to the month. Predetermined exposure is 2014 external debt stocks as a share of GNI from the World Bank, transformed as $\log(1 + x)$ and standardized in the primary specification. All regressions include country and year-month fixed effects, three lags of the AERI change and three lags of the shock–exposure interaction. Standard errors are two-way clustered by country and month. Reported p-values use a finite-cluster t reference with degrees of freedom equal to the number of African-country clusters minus one. Restricted wild-cluster-bootstrap p-values for the primary month-four and month-nine coefficients are 0.042 and 0.034, respectively. In the leave-one-country-out exercise, 28 of 29 month-four estimates and 24 of 29 month-nine estimates also satisfy $p < 0.05$.

significance across all six variants. The result also remains when the time sample is restricted to FOMC announcement months.[6] In the leave-one-country-out calculations, the month-four coefficient ranges from 0.214 to 0.308, with all 29 estimates significant at 10 percent and 28 at 5 percent. At month nine it ranges from 0.282 to 0.455; all 29 are significant at 10 percent and 24 at 5 percent.

The falsification tests do not show the same pattern. Current monetary-policy surprises have little relationship with AERI changes dated one to six months earlier; finite-cluster p-values range from 0.156 to 0.807. The central-bank-information component of the same announcements produces no comparable debt-exposure pattern at months four and nine ($p = 0.153$ and $p = 0.125$). The joint future-shock interaction placebo yields $F(6, 28) = 1.884$ with $p = 0.119$, and the restricted wild-cluster bootstrap gives $p = 0.221$.[7] The 2014 debt ratio is predetermined but observational. It may be correlated with exchange-rate regimes, trade openness, institutional quality and other persistent country characteristics. The coefficient measures how the response to the external shock varies with pre-existing debt exposure; it does not by itself isolate a causal debt-transmission channel.

5 Conclusion

The paper constructs two news-based measures of economic risk for 54 African economies over 2015–2025. AERI aggregates graded article-level scores, the pillar indices retain the economic domain attached to those scores, and ARS provides a bounded summary built from the monthly risk signal. The more than 21 million retained article records show both common episodes and different country histories. The macroeconomic results are also directional rather than uniform: inflation-risk readings precede movements in headline inflation, exchange-rate depreciation is followed by higher AERI, stronger real activity is followed by lower AERI, and an externally identified U.S. monetary-policy tightening is followed by a larger AERI increase in African economies with greater predetermined external-debt exposure.

AERI, the pillar indices and ARS answer different questions. AERI records the amount of scored economic-risk content relative to the news flow. The pillar indices show which domains account for it. ARS indicates whether the monthly signal is narrow or spread across more of the risk dimensions used in the score. A sharp AERI increase with little movement in ARS therefore describes a different episode from one in which both measures

[6] This robustness specification changes the time observations only; the cross-section remains the same set of African economies.

[7] The corresponding large-sample Wald statistic is $\chi^2(6) = 11.30$ with asymptotic $p = 0.079$. With 29 country clusters, the finite-cluster and restricted wild-cluster-bootstrap references provide the more appropriate finite-sample benchmarks.

rise.[8]

Cross-country comparison also has limits. Economic structure, external exposure, media depth, language and the availability of conventional high-frequency data differ across the sample. The common scoring rule fixes the definition of the measure, but country interpretation still depends on the source mix and the composition of the pillar scores.[9] Appendix B finds little within-country association between AERI and retained article volume. Newspaper archives can therefore add timely information where conventional data arrive slowly, but the index remains a measure of the observed news environment rather than a substitute for country-specific macroeconomic analysis.

5.1 Limitations

The measures are bounded by what appears in the retained news archive. AERI, the pillar indices and ARS describe reported economic risk in that information environment; they do not measure realised losses or crisis probabilities. Cross-pillar comparisons have the same limitation. A high financial-sector pillar mean, for example, means that financial-sector concerns are more prominent in the corpus, reflecting both underlying conditions and reporting practices.

Media coverage creates a separate source of uncertainty. Publication volume, editorial priorities, source continuity and language coverage differ across countries and over time. In the eligible sample, AERI has little within-country association with log article volume, so archive depth alone does not explain the index. That result does not establish source representativeness. The production dictionaries explicitly cover English, French and Portuguese, and reporting in other languages may be captured less completely. Exact-text deduplication removes identical reprints but leaves edited versions, translations and related stories as distinct records. The reported deduplication sensitivity applies to the descriptive series; the PVAR and external-shock specifications continue to use the original benchmark vintage.

The PVAR coefficients are conditional averages for the selected countries and depend on a researcher-imposed contemporaneous ordering. The main sign pattern appears under both the forward and feedback orderings, but the recursive system is not a structural causal design. We use it to describe dynamic relationships and rely on the external-shock specification for the separate identification exercise.

The monetary-policy shock is externally identified, but 2014 external-debt exposure is observational and may reflect persistent country differences. In the 29-country sample, we therefore read the result as an exposure-sensitive AERI response, not a causal debt-

[8] Joint movements in AERI, ARS and the pillar indices distinguish concentrated episodes from broader cross-domain shifts and guide follow-up analysis.

[9] The same headline value can reflect different pillar compositions, so country interpretation should use the pillar profile and coverage diagnostics.

transmission estimate.

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A Data Collection Architecture

The collection and measurement sequence is: country-site registry; monthly retrieval; text normalisation; relevance classification; severity and textual emphasis weighting; country-month aggregation; pillar diagnostics; and ARS construction. A failed source month is treated as a gap in the observation system, while a retained month with no matched economic content is recorded as a measured zero. The replication extension adds exact-text comparison within country-months, including across outlets, and retains duplicate counts, representative examples and file fingerprints. This step addresses identical article bodies; broader questions of story independence and media representativeness remain part of the measurement limitations discussed in the main text.

The index vintage spans 131 or 132 reported months per country within 2015–2025. Annual country values are available from the authors upon request. Descriptive comparisons generally require at least 100 articles per month. The PVAR sample is defined separately: it starts from the 36-country correlation-screened set, retains country-months with complete model inputs and the required lag structure, and requires at least 24 usable observations per country. The descriptive 100-article threshold is not imposed on PVAR estimation. Formal PCA derivations, detailed pillar descriptions and expanded robustness results are reported in the online supplement.

B Archive-Depth Diagnostic

Table B.1 examines whether AERI moves mechanically with the number of retained articles after persistent country differences are removed. The first specification includes country fixed effects; the second adds year-month fixed effects. Both use country-clustered standard errors and the 6,488 country-months that satisfy the principal coverage threshold.

Table B.1: AERI and within-country news-archive depth

Specification	Coefficient	Std. error	p-value	Observations
Country fixed effects	1.141	2.978	0.702	6,488
Country and year-month fixed effects	-1.474	3.153	0.640	6,488

Notes: The dependent variable is AERI and the regressor is the natural logarithm of retained article records in the country-month. The sample contains the 6,488 country-months with at least 100 retained articles, covering 53 countries; Guinea-Bissau has no month meeting the threshold in this vintage. Standard errors are clustered by country. The within-country correlation between AERI and log article volume is 0.035. The second specification also absorbs year-month fixed effects.

C External-Identification Diagnostics

Table C.1 reports falsification and small-cluster inference checks for the externally identified monetary-policy exposure design. The joint future-shock placebo includes the contemporaneous shock–exposure interaction as a control. Pre-shock outcome placebos are estimated without lagged outcome controls that would mechanically overlap the placebo dependent variable. Complete horizon paths, robustness results, leave-one-country-out estimates, finite-cluster calculations, wild-cluster-bootstrap outputs and the underlying monthly shock and exposure files are available from the authors on request and are summarised further in the online supplement.

Table C.1: External-identification diagnostics and falsification checks

Check	Statistic / range	Interpretation
Pre-shock AERI-change placebos (1–6 months)	$p \in [0.156, 0.807]$	None significant at 10 percent
CBI shock × exposure, month 4	$p = 0.153$	p-value above 0.10
CBI shock × exposure, month 9	$p = 0.125$	p-value above 0.10
Joint future-shock interaction placebo	$\chi^2(6) = 11.30$, asymptotic $p = 0.079$	Large-sample reference only
Finite-cluster joint reference	$F(6, 28) = 1.884$, $p = 0.119$	p-value above 0.10
Restricted wild-cluster bootstrap	$p = 0.221$	p-value above 0.10

Notes: The joint future-shock placebo includes the contemporaneous monetary-policy-shock–exposure interaction as a control. The asymptotic Wald reference is reported for transparency, but the preferred joint inference recognizes that the identifying cross-section contains 29 African-country clusters. The finite-cluster F reference uses 28 denominator degrees of freedom; the restricted wild-cluster bootstrap uses 999 Rademacher replications by African country. The pre-shock outcome placebos are estimated without lagged dependent-variable controls that would mechanically span the placebo outcome.

D PVAR Sample and Estimation Details

D.1 Macroeconomic Data and Pooled Panel Estimation

The empirical assessment relates monthly news-based risk to headline inflation, exchange-rate depreciation and monthly GDP growth. Its purpose is to test whether the measured domains display economically interpretable dynamics. ARS calibration is completed independently of these macroeconomic outcomes, preserving a clean separation between index construction and subsequent validation.

D.1.1 Monthly variables and sample construction

Headline inflation, π_{ct}, is the supplied monthly observation of the inflation rate. The exchange rate, e_{ct}, is expressed as local-currency units per US dollar. Monthly GDP, G_{ct}, is the supplied purchasing-power-parity GDP series. Depreciation and monthly GDP

growth are defined as

$$d_{ct} = 100[\log e_{ct} - \log e_{c,t-1}], \qquad g_{ct} = 100[\log G_{ct} - \log G_{c,t-1}]. \tag{D.1}$$

Thus a positive d_{ct} denotes depreciation, whereas a positive g_{ct} denotes monthly GDP growth. Headline inflation retains the definition supplied in the macroeconomic panel, while depreciation and GDP growth are the log-difference transformations above. The sample-specific winsorisation bounds are reported below.

The main validation specification uses one news-based risk measure at a time—the inflation-risk pillar, the FX and external-risk pillar, AERI or ARS—together with inflation, depreciation and monthly GDP growth. The estimation sample is distinct from the descriptive sample defined by the 100-article threshold. It begins with the 36-country union selected by the prior correlation screen, then retains country-months with the risk measure, the required macroeconomic inputs and the full lag structure, subject to at least 24 usable observations per country.[10] The common GDP-augmented panel contains 4,286 observations for 36 countries and runs from May 2015 to December 2025. The 100-article rule continues to govern the principal descriptive country comparisons and ARS calibration; it is not imposed on the PVAR estimation panel. This distinction explains why a country can appear in the PVAR sample even when few or no months satisfy the descriptive coverage threshold. Using the direct pillar indices in this specification keeps the economic interpretation aligned with the measurement framework in the main text. Supporting transformations and country-level estimation diagnostics are available from the authors on request.

Risk innovations are put on a comparable within-country scale. Let $\bar{r}_c$ be the country mean in the supplied scaling panel and $s_{r,w}$ the sample standard deviation of $r_{ct} - \bar{r}_c$ pooled across that panel. We use $\widetilde{r}_{ct} = r_{ct}/s_{r,w}$; country effects subsequently absorb level differences. A one-unit risk innovation therefore represents one pooled within-country standard deviation, rather than one raw index point. Scaling parameters are fixed for the reported estimates and bootstrap exercises.

D.1.2 The pooled dynamic system

For m endogenous variables, the maintained model is

$$y_{ct} = \alpha_c + \lambda_t + \sum_{\ell=1}^{p} B_\ell y_{c,t-\ell} + \varepsilon_{ct}, \qquad E(\varepsilon_{ct}\varepsilon_{ct}') = \Sigma. \tag{D.2}$$

[10]The estimation panel winsorises the supplied macroeconomic transformations at pooled first and ninety-ninth percentiles. In the GDP-augmented sample the retained bounds are $[-2.565, 50.934]$ for inflation, $[-5.736, 14.926]$ for depreciation and $[-0.819, 1.506]$ for monthly GDP growth.

Here α_c captures persistent country differences, and λ_t is a full year-month effect that absorbs shocks common to countries in a given month. The year-month effects therefore absorb common calendar-time variation rather than recurring month-of-year seasonality alone. The slope matrices B_ℓ are pooled across countries. In the three-variable system, $y_{ct} = (\widetilde{r}_{ct}, \pi_{ct}, d_{ct})'$. In the GDP-augmented system, $y_{ct} = (\widetilde{r}_{ct}, \pi_{ct}, d_{ct}, g_{ct})'$. In particular, the inflation equation in the latter system is

$$\pi_{ct} = \alpha_c^{\pi} + \lambda_t^{\pi} + \sum_{\ell=1}^{p} \big(b_{\pi r,\ell} \widetilde{r}_{c,t-\ell} + b_{\pi\pi,\ell} \pi_{c,t-\ell} + b_{\pi d,\ell} d_{c,t-\ell} + b_{\pi g,\ell} g_{c,t-\ell} \big) + \varepsilon_{ct}^{\pi}. \tag{D.3}$$

The risk, depreciation and GDP equations have the same lag structure with equation-specific coefficients. Each outcome can therefore respond to its own history and to the histories of the other variables. Pooling provides a common dynamic description conditional on fixed effects. The multivariate panel structure follows the empirical framework in Love and Zicchino (2006) and Abrigo and Love (2016), implemented here with within-transformed pooled least squares.

Let X stack all mp lagged regressors and Y stack the contemporaneous outcomes on the common estimation sample. Alternating country and year-month projections yield $\ddot{X}$ and $\ddot{Y}$. The pooled estimator and residual covariance are

$$\widehat{B} = (\ddot{X}'\ddot{X})^{+}\ddot{X}'\ddot{Y}, \quad \widehat{E} = \ddot{Y} - \ddot{X}\widehat{B}, \quad \widehat{\Sigma} = \frac{\widehat{E}'\widehat{E}}{n - mp}. \tag{D.4}$$

The superscript $+$ denotes the Moore–Penrose inverse; the covariance denominator is the implemented lag-regressor adjustment. A common maximum-lag sample is used when comparing candidate orders, preventing changes in the available observations from driving lag choice. Among stable candidates with $p \in \{1, 2, 3\}$, the criterion is

$$\mathrm{BIC}(p) = \log |\widehat{\Sigma}_p| + \frac{\log n}{n} m^2 p. \tag{D.5}$$

Stability requires the maximum absolute companion-matrix eigenvalue to lie below the numerical threshold of 0.999. All reported specifications select three lags. Table D.1 reports observations, country counts, roots and accepted bootstrap draws. Fixed-effects estimation with lagged outcomes can retain finite-time bias (Nickell, 1981); the long monthly dimension reduces some finite-sample concerns but leaves this source of bias relevant to interpretation.

D.1.3 Identification and impulse responses

Let $\widehat{\Sigma} = LL'$ with L lower triangular. Forward systems order risk first, followed by inflation, depreciation and monthly GDP growth where included. This permits a risk innovation to affect the macroeconomic variables within the month, while assigning contemporaneous feedback into risk to subsequent periods. For reverse responses, the order is inflation, depreciation, monthly GDP growth and AERI. The two orderings answer complementary recursive questions about forward risk propagation and macroeconomic feedback.

The moving-average matrices and normalised responses are

$$\Phi_0 = I_m, \qquad \Phi_h = \sum_{\ell=1}^{\min(p,h)} \widehat{B}_\ell \Phi_{h-\ell}, \tag{D.6}$$

$$\theta_{kj}(h) = \frac{e_k' \Phi_h L e_j}{e_j' L e_j}, \qquad h = 0, \ldots, 12, \tag{D.7}$$

where e_j is the jth coordinate vector, distinct from the exchange-rate notation in equation (D.1). The denominator normalises the innovation to a one-unit contemporaneous increase in the selected variable. For forward risk innovations, that unit is one within-country standard deviation. For reverse macroeconomic innovations, it is one percentage point. This convention makes the reported response magnitudes explicit while preserving the natural units used for the macroeconomic innovations.

D.1.4 Inference and interpretation

Uncertainty bands are constructed by resampling countries with replacement, retaining each selected country's complete time history, and re-estimating the fixed-effects system. The lag order, measurement transformations and fixed ARS calibration remain unchanged across bootstrap replications. Four hundred accepted stable replications provide the fifth and ninety-fifth percentiles at each horizon. These are pointwise 90 percent intervals, so each band describes uncertainty at a given horizon. Simultaneous inference for the full response path or for the location of its population peak would require a different procedure.

Joint lag tests complement the IRFs. For example, the inflation equation tests

$$H_0 : b_{\pi r,1} = \cdots = b_{\pi r,p} = 0, \qquad W = (R\widehat{\beta})'[R\widehat{V}R']^{+}(R\widehat{\beta}), \tag{D.8}$$

using country-cluster covariance $\widehat{V}$ and a χ^2_p reference distribution. The exact covariance calculation is given in Appendix D. Rejection indicates incremental information in the included lagged variable conditional on the other lags and fixed effects. The recursive response additionally reflects contemporaneous residual covariance and propagation through the full system, so the tables report both forms of evidence.

The 36-country GDP extension is based on the union of countries meeting the pre-specified inflation or exchange-rate correlation screen and contains 4,286 common-sample observations.[11] Selection is part of the design, and inference is conditional on it. Table D.1 reports the sample sizes, lag choices, stability roots and accepted bootstrap draws. The GDP-augmented roots lie between 0.919 and 0.923.

Table D.1: Pooled PVAR samples and stability diagnostics

System	Variables	Countries	Obs.	Lags	Max. root	Draws
Inflation risk + GDP	4	36	4,286	3	0.9228	400/400
FX/external risk + GDP	4	36	4,286	3	0.9222	400/400
AERI + GDP	4	36	4,286	3	0.9198	400/400
ARS + GDP	4	36	4,286	3	0.9190	400/400

Notes: All samples span May 2015–December 2025 and use the same 36-country correlation-selected panel. The PVAR sample is formed from complete model inputs and lags with at least 24 usable observations per country; the 100-article rule used for descriptive country comparisons is not imposed here. Draws report accepted/attempted country-cluster replications. A companion root below one indicates fitted dynamic stability; identification is assessed separately through the stated recursive and external-shock designs.

E Supplementary Country Outputs

Appendix Table E.1 reports monthly AERI summary statistics for the illustrative ten-country group used in the annual comparison. The online supplement contains the full 54-country AERI/ARS summary, the two multi-country monthly path figures, detailed PVAR estimation material, the expanded exact-text deduplication results and additional external-identification robustness output.

[11]The panel comprises Algeria, Angola, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Comoros, Democratic Republic of Congo, Egypt, Ethiopia, Gambia, Ghana, Guinea, Guinea-Bissau, Ivory Coast, Lesotho, Liberia, Libya, Malawi, Mali, Mauritius, Morocco, Mozambique, Namibia, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, South Africa, Tanzania, Tunisia, Uganda, Zambia and Zimbabwe.

Table E.1: Country-level AERI: descriptive comparisons, 2015–2025

Country	Months	Mean	SD	Peak	Peak month
Nigeria	132	73.04	8.75	98.43	2024-02
Ghana	132	57.11	10.47	85.70	2025-03
Algeria	132	65.02	6.34	83.44	2020-04
Morocco	132	22.67	7.54	37.66	2025-10
Egypt	132	40.55	11.29	97.11	2025-12
Ethiopia	132	59.96	17.50	127.83	2019-02
South Africa	132	42.58	7.61	79.22	2020-04
Kenya	132	38.85	7.02	71.03	2025-12
Angola	132	53.45	8.60	85.67	2019-10
Ivory Coast	132	30.22	5.28	46.87	2025-04

Notes: Monthly AERI using the article-score definition in equation (5). Only months with at least 100 retained articles are included. Means and sample standard deviations give equal weight to eligible months; they differ from article-weighted annual AERI. Peak denotes the maximum eligible monthly observation, with the earliest month reported in a tie. Countries are the same illustrative group as Figure 4; PVAR sample selection is conducted separately.